Name of Student: Rikisha Bhattacharya	**PRN: 2507122155**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No:	
TITLE: JDBC DriverManager, Connection, and Statement	

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).

Learning Objectives

To understand and use the core JDBC classes for database connectivity.

1. Create a program to connect Java with database and display database connection status.
2. Create a Java program to connect with a database and display the connection status. After successful connection, show a message indicating that the Student database is connected successfully.

Programme

Welcome | databaseConnectionStatus.java

databaseConnectionStatus.java > databaseConnectionStatus > main(String[])

```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

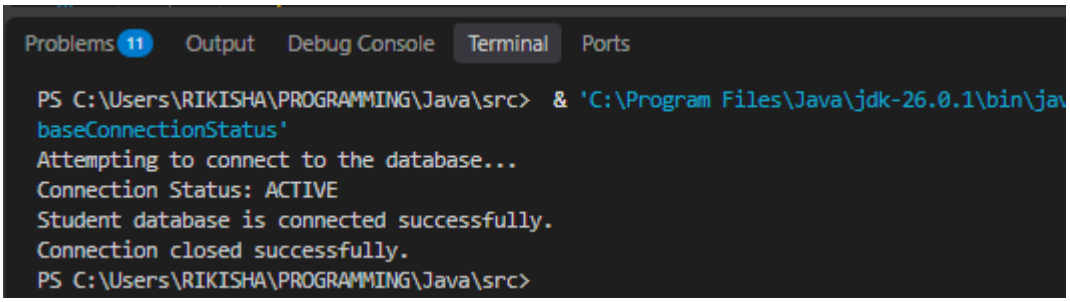
public class databaseConnectionStatus {
    Run | Debug
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/rikishab";
        String user = "root";
        String password = "YourNewPassword123!";

        System.out.println(x: "Attempting to connect to the database...");

        try {
            Class.forName(className: "com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(url, user, password);

            if (conn != null && !conn.isClosed()) {
                System.out.println(x: "Connection Status: ACTIVE");
                System.out.println(x: "Student database is connected successfully.");
                conn.close();
                System.out.println(x: "Connection closed successfully.");
            } else {
                System.out.println(x: "Connection Status: FAILED");
            }
        } catch (ClassNotFoundException e) {
            System.out.println(x: "JDBC Driver not found.");
            e.printStackTrace();
        } catch (SQLException e) {
            System.out.println(x: "Database Connection Failed.");
            e.printStackTrace();
        }
    }
}
```

Output



Conclusion:

Thus, the JDBC classes and interfaces, including DriverManager, Connection, and Statement, were explored to understand database connectivity in Java. The programs successfully demonstrated establishing a database connection and displaying its connection status.